



UNIVERSITI MALAYA

WIA2006 | SYSTEMS ANALYSIS AND DESIGN

**PROJECT REPORT
SEMESTER 1, SESSION 2024/2025**

TITLE: sapU (Smart AI-Powered Carpool System)

Group: OCC 1

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Summary of Project:

The sapU (Smart AI-Powered Carpool System) project was developed to address transportation challenges faced by university students, particularly issues related to affordability, safety, and inefficiency in existing commuting options. Many students rely on costly commercial e-hailing services or unregulated informal ride-sharing platforms, which often lack proper verification, coordination, and safety mechanisms.

This project aims to bridge the gap between these two extremes by providing a structured, student-centered carpooling solution. Using a structured Systems Analysis and Design approach, the project followed the Software Development Life Cycle (SDLC) with Agile methodology to ensure flexibility and continuous improvement. Comprehensive fact-finding techniques, including field observations, interviews, and questionnaire surveys, were conducted to identify real student needs. The findings highlighted high transportation costs, safety concerns, and operational inefficiencies, which directly influenced the system's functional and non-functional requirements.

The sapU system integrates AI-driven route matching, a built-in price negotiation feature, real-time GPS tracking, and strict user verification to enhance efficiency, affordability, and trust. The system architecture adopts a client-server model supported by a cloud-based database to manage users, rides, payments, and safety events. Dedicated interfaces were designed for passengers, drivers, and administrators to ensure usability and effective monitoring. Evaluation and testing were carried out through multiple testing techniques, including unit, integration, system, and user acceptance testing, to validate system reliability and usability.

Overall, the sapU project demonstrates the potential of an AI-powered, secure, and sustainable carpooling platform to improve campus mobility, reduce transportation costs, and promote safer commuting among university students.

1. INTRODUCTION

The university environment presents unique and complex logistical challenges regarding student mobility, where students must frequently travel between residential colleges, faculty buildings, and nearby urban centers. Currently, students on campus face significant difficulties in finding affordable and reliable transportation that aligns with their hectic academic schedules. This problem is observed daily as the student population navigates limited campus shuttle availability and peak traffic periods, which often leaves them with few viable options. Consequently, many students are forced to rely on expensive commercial e-hailing services or resort to informal ride requests through unregulated social media channels like WhatsApp and Telegram. These informal environments lack the structure, security, and technological integration necessary for a consistent and professional commuting experience, leading to high levels of uncertainty for both passengers and drivers.

The **sapU (Smart AI-Powered Carpool System)** project is a critical intervention for the university community. This project is important because it directly addresses the financial

and safety concerns of the student body. The benefits of this system are:

- Students benefit from significantly reduced travel costs and shorter waiting times through a negotiation system and efficient route matching.
- Student-drivers can generate side income and offset vehicle maintenance costs while maintaining a stable work-study-life balance.
- By consolidating multiple passengers into a single vehicle, the system supports sustainability by reducing overall traffic volume and carbon emissions.

Improvement is necessary because existing methods; both formal and informal, fail to meet the specific economic and safety needs of the student's demographic.

1.1 Statement of Problem

There is currently a significant gap between professional e-hailing services and informal ride-sharing methods that the sapU system aims to fill. While existing e-hailing apps are secure, they are often not student-friendly due to higher fares and limited availability within campus grounds. Conversely, manual ridesharing through social media is unregulated and lacks essential features such as GPS tracking and reliable matching algorithms, making it inherently unsafe and disorganized.

sapU fills this void by providing a specialized mobile-based carpool platform that integrates AI route-matching with strict user verification, offering a safe and affordable alternative that existing systems do not currently provide.

1.2 Project Objectives

To address the identified gaps and improve the current transportation situation, the project will pursue the following objectives:

- Utilize AI route-matching algorithms to connect passengers and drivers with similar routes, thereby reducing waiting times and travel costs.
- Implement a robust safety framework including user verification, real-time GPS tracking, and AI-based monitoring to build user trust.
- Facilitate a pricing system that allows students to bargain for affordable rides and provides drivers with a platform for side income.
- Reduce the campus carbon footprint by optimizing car occupancy and decreasing traffic congestion.

1.3 Project Scope

The sapU system is designed with specific boundaries to ensure feasibility within the 11-week project timeline.

Included in Scope:

- **Target Users:** Exclusively university students.
- **Functionality:** Development of a mobile-based platform featuring AI route-matching, user verification, rating system, and an admin dashboard.
- **Geographical Focus:** The system is limited to short and medium distance rides within the campus grounds and nearby urban areas.

Exclusions and Constraints:

- The system will not operate as a commercial e-hailing business; it is strictly a carpooling platform.
- The project is limited by a fixed budget of RM 4,900.00 and depends on the accuracy of third-party GPS data.

2. PROJECT ROLES

Table 1. Project Roles

Position	Name	Tasks
Project Manager	Najah Safwah	• Plans project timeline, tasks, and milestones • Manages team communication. • Tracks progress, risks, and deadlines • Coordinates meetings & ensures everyone stays on scope.
System Analyst	Fatin Nurliyana	• Gathers requirements from stakeholders. • Creates system specifications. • Designs workflows • Ensures the system solves the correct problem.
UI/UX Designer	Nurul Hannah	• Designs the user interface. • Plans the user experience and navigation flow. • Ensures accessibility and usability. • Prepares design assets for developers.
Frontend Developer	Nurin Allya Insyira	• Implements the UI/UX design into actual screens. • Develops interactive features. • Connects front-end to back-end APIs. • Ensures responsive design.
Backend Developer	Nur Sahira Batrisyia	• Builds the server, database, and core logic. • Handles authentication, data processing, and APIs. • Ensures security, performance, and data accuracy. • Integrates external services if needed.
Quality Assurance	Anis Atirah	• Tests the system for bugs and issues. • Ensures all requirements are met. • Verifies system before deployment.

3. BUDGET

Table 2. Estimated budget

Item	Amount	Unit	Sub-Total (RM)
Project Management	60 hours @ RM25/hour	60 hours	1500.00
UI/UX design	Wireframes + mockups	40 hours	1200.00
Front-end / No code development	Prototype build	30.00	1800.00
Quality assurance / Testing	Test cases and bug fixes	20.00	400.00
Total			(RM4900.00)

4. PROJECT TIMELINE

Figure 1. Gantt chart of sapU project timeline

SAPU Project Timeline

PROJECT TITLE		SAPU SYSTEM				COMPANY NAME		UDANG MAYO SDN. BHD.							
PROJECT MANAGER		NAJAH SAFWAH BINTI MONARIZZAL				DATE		19/10/2025							
NO.	TASK TITLE	TASK OWNER	START DATE	DUE DATE	STATUS	PHASE ONE		PHASE TWO		PHASE THREE		PHASE FOUR			PHASE FOUR
						W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
1	Project Conception and Initiation														
1.1	Project Charter	Everyone	13/10	19/10	Completed										
1.2	Objectives & Scopes	Everyone	13/10	26/10	Completed										
1.3	Target Audience	Everyone	20/10	26/10	Completed										
1.4	Role Assignment	Everyone	20/10	26/10	Completed										
2	Project Definition and Planning														
2.1	Problem Identification	Everyone	27/10	2/11	Completed										
2.2	Budget	Everyone	27/10	2/11	Completed										
2.3	Requirements & User Needs	Everyone - System Analyst	27/10	9/11	Completed										
2.4	Feasibility Analysis	System Analyst	3/11	9/11	Completed										
2.5	Methods	Everyone	3/11	9/11	Completed										
MILESTONE 1: PLANNING & FEASIBILITY APPROVED															
3	Project Development I														
3.1	UI Design	UI/UX Designer	10/11	16/11	Completed										
3.2	Prototype Building	Frontend Developer	10/11	23/11	Completed										
4	Project Development II														
4.1	App Building	Frontend Developer	24/11	14/12	Completed										
4.2	Feature Integration	Backend Developer - Frontend Developer	1/12	14/12	Completed										
4.3	Database Design	Backend Developer	1/12	14/12	Completed										
MILESTONE 2: SYSTEM DEVELOPMENT COMPLETED															
4.4	Internal Testing	Quality Assurance	15/12	20/12	Completed										
4.5	User Testing & Feedback	Everyone - System Analyst - Quality Assurance	22/12	20/12	Completed										
4.6	Issues & Improvements	Quality Assurance - Frontend Developer - Backend Developer	22/12	20/12	Completed										
5	Project Launch & Presentation														
5.1	Final Deployment	Everyone	29/12	4/1	Completed										
MILESTONE 3: PROJECT DELIVERED															

Table 3. Description of each Phase

PHASE 1 (Initiation)	WEEK 1 - WEEK 2 - Planning and formation of development team - Defining project objectives - Selecting target audience - Assigning project manager, roles of each team member, and project budget
PHASE 2 (Planning)	WEEK 3 - WEEK 4 - Establish project problem statement - Discuss on system requirements to overcome the problem stated - Identify project economic, technical, operation, legal and contractual, political, schedule, timeline, and resource analysis - Identify methods of collecting system requirements
PHASE 3 (Design)	WEEK 5 - WEEK 6 - Building system prototype with functioning fundamental features - Create system's UI/UX design
PHASE 4 (Implementation)	WEEK 7 - WEEK 10 - Build core app functionality - Develop the AI navigation feature - Apply the real-time database for prices and drivers - Test out the system through internal, alpha and beta testing - Gain feedback from users - Implement improvements derived from user feedback
PHASE 5 (Release)	WEEK 11 Broadcast the system for public use

MILESTONES

- **Planning and Feasibility Approval (End of Week 4):** Completion of initial project planning, team formation, requirement gathering, and feasibility assessment to ensure the project is viable and objectives are well-defined.
- **System Development Completion (End of Week 9):** Finalization of core system functionalities, including AI route-matching, user interface implementation, database integration, and initial internal testing.
- **Project Delivery (Week 11):** Deployment of the fully developed sapU system for public use and official release to the target student community.

5. PLANNING

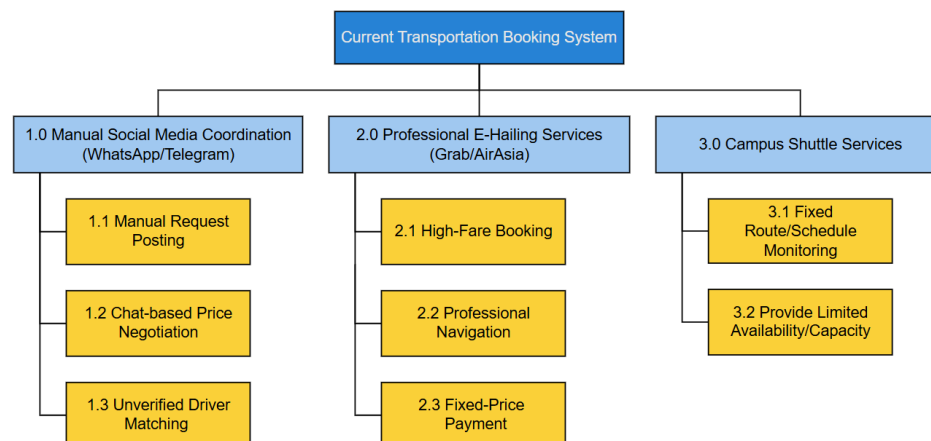
5.1 Fact Finding Conducted

To accurately capture the requirements for the sapU system, the development team conducted a multi-layered fact-finding process to bridge the gap between current student struggles and a functional technological solution. This research was essential to ensure the system addresses real-world campus mobility issues. The primary methods included:

- **Field Observations:** The team monitored high-traffic areas on campus, such as residential college bus stops and faculty drop-off points, during peak hours. This allowed the team to observe the long wait times and the visible frustration of students dealing with limited campus shuttle capacity.
- **Digital Environment Analysis:** An analysis of informal ride-sharing groups on platforms like Telegram and WhatsApp was conducted. This revealed a disorganized flow of information where students manually post requests, leading to "redundant orders" where multiple drivers respond to one student, or many requests go unanswered due to a lack of coordination.
- **Student Interviews:** The team conducted qualitative interviews with a diverse group of students to understand their financial and safety concerns. Key insights from these interviews showed that many students suffer from financial instability due to high commercial e-hailing fares and feel anxious about the lack of user verification in informal chat groups.
- **Questionnaire Survey:** To complement the qualitative insights from interviews, the team distributed a structured online questionnaire between October and November 2025, receiving 156 responses from students aged 20 to 24. This quantitative method allowed the team to measure the scale of the transportation crisis across different academic years and living arrangements.

5.2 Existing Systems FDD

Figure 2. FDD of Current Transportation Booking System



5.3 Limitations

The analysis of the current situation revealed several critical limitations that hinder student mobility and safety:

- **Safety and Security Risks:** There is a total lack of user verification, real-time GPS tracking, and safety monitoring in current informal groups.
- **Financial Strain:** Professional e-hailing fares are often too high for a student's budget, and there is no formalized way to bargain for lower prices.
- **Operational Inefficiency:** The manual process is complex, redundant, and disorganized, leading to wasted time as users wait for matches or deal with multiple drivers for a single trip.
- **Unreliability:** Informal ridesharing is inconsistent; there is no guarantee a driver will be available for a specific route or timing.

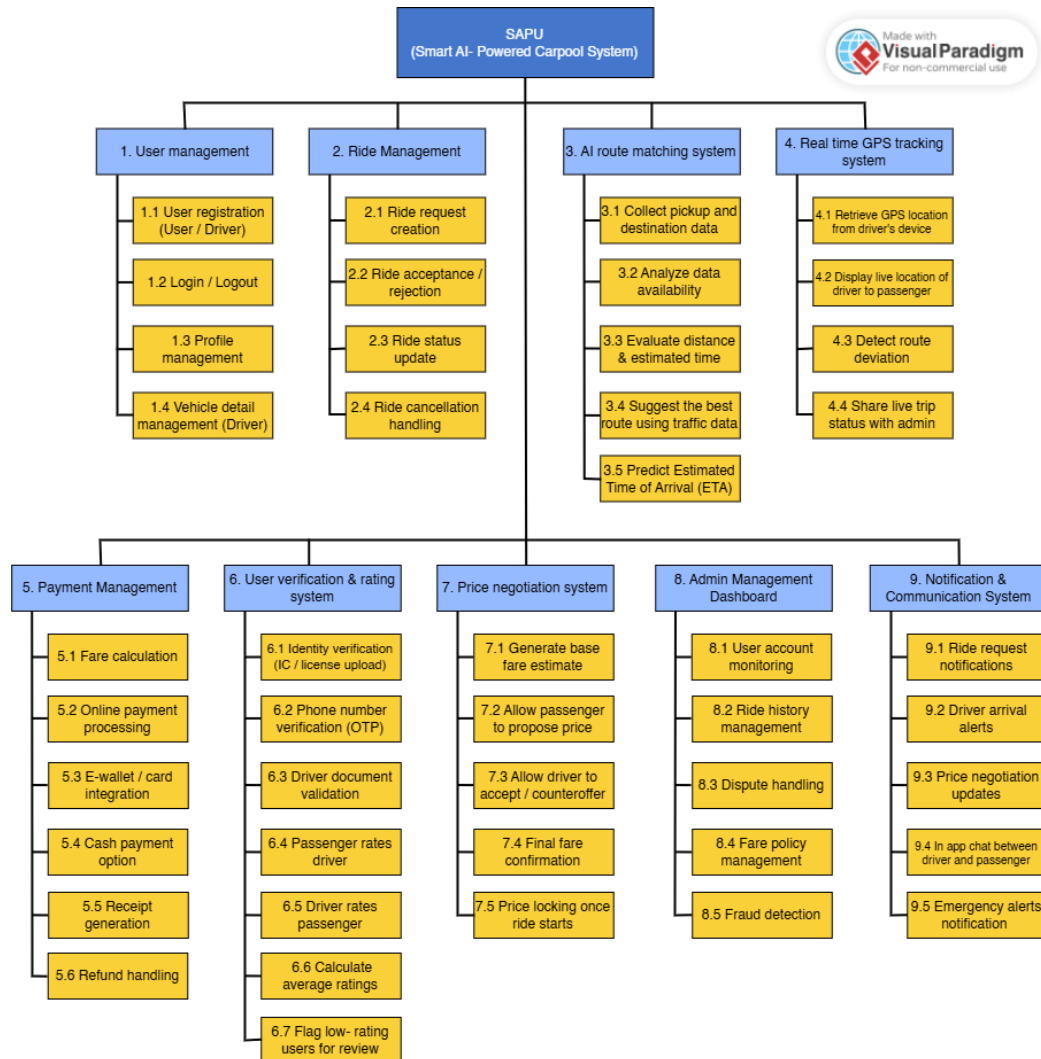
5.4 Recommendations

To overcome these limitations, the following features are suggested for the proposed sapU system:

- **AI-Powered Matching:** Implement an AI route-matching algorithm to automatically connect drivers and passengers with similar routes and timing to ensure efficiency.
- **Formalized Bargaining System:** Integrate a feature that allows students to negotiate and agree on affordable ride prices within the app.
- **Safety Infrastructure:** Include mandatory user verification, real-time GPS tracking, and a rating analysis system to build user trust.
- **Centralized Mobile Interface:** Transition from fragmented chat groups to a structured platform to avoid redundant orders and save time.

5.5 Proposed System FDD

Figure 3. FDD of sapU (Smart AI-Powered Carpool System)



6. ANALYSIS

6.1 User Research

6.1.1 Questionnaire Results

Previous studies have shown that university students often face affordability and accessibility challenges in daily commuting [1], [2]. To identify transportation challenges faced by students at the University of Malaya (UM), a questionnaire-based survey was conducted between October and November 2025. The survey aimed to gather information on student demographics, accommodation status, and daily commuting practices in order to justify the need for a structured campus carpooling system.

A total of 66 students aged between 20 and 24 years participated in the survey. The questionnaire was distributed online through university email channels and student social media groups to ensure broad participation across faculties and academic years. Responses were fairly balanced, with Year 2 students forming the largest group at 31%.

The survey revealed that 58% of respondents live off-campus, while 42% reside within campus grounds. Only 35% of students reported owning personal transportation, indicating a strong reliance on alternative commuting methods among the majority of the student population.

Among students with personal transport, motorcycles were the most commonly used mode (42%), followed by cars (30%). Students without personal transport relied primarily on e-hailing services such as Grab (41%), followed by public transport including buses and LRT (28%), and informal carpooling arrangements (19%).

Several critical issues emerged from the findings. Students using e-hailing services reported spending between RM150 and RM300 per month on transportation, representing a significant financial burden. Informal carpooling users highlighted challenges such as inconsistent schedules, lack of trust, poor coordination, and uncomfortable payment arrangements. Additionally, students living in areas with limited public transport connectivity experienced commute times of up to 90 minutes.

Importantly, 73% of students who own vehicles indicated that they usually travel alone and would be willing to offer rides if a secure and organized platform were available. Safety concerns were particularly prominent among female students, emphasizing the importance of user verification and real-time monitoring. These findings directly influenced the functional and safety-oriented design of the sapU system.

The survey findings indicate a strong dependence on informal and costly transportation methods, highlighting the lack of a centralized and secure carpooling platform for students. These results justify the need for a system that supports ride matching, cost negotiation, safety monitoring.

6.1.2 User Personas

Based on the survey results and follow-up discussions, three representative user personas were developed to guide system design.

Persona 1: Sarah – The Budget-Conscious Passenger

Sarah is a third-year student living off-campus without personal transport. She relies heavily on e-hailing services, which consume a large portion of her monthly allowance. Her primary goals are to reduce transportation costs, ensure personal safety, and secure reliable daily rides. sapU addresses her needs through verified drivers, affordable negotiated pricing, and real-time trip tracking.

Persona 2: Ahmad – The Student Driver

Ahmad is a final-year student who drives to campus daily. Although he frequently travels alone, he faces high fuel and maintenance costs. His objectives include earning additional income, managing ride requests efficiently, and maintaining flexibility. sapU enables Ahmad to list rides along his existing route, manage bookings digitally, and build trust through a rating system.

Persona 3: Dr. Lim – Campus Administrator

Dr. Lim represents university administrators responsible for student safety and transport oversight. His challenges include limited visibility into informal transportation arrangements and delayed incident response. The sapU admin dashboard supports his role through real-time monitoring, automated alerts, and comprehensive reporting tools.

6.1.3 Use Cases

Use Case 1: Passenger Requests and Books a Ride

A passenger enters pickup and destination details along with the desired time. The system matches the request with suitable drivers based on route similarity, availability, and ratings. Once a driver accepts, both parties receive confirmation, driver information, and live tracking access.

Use Case 2: Price Negotiation Between Passenger and Driver

The system generates an initial fare based on distance and traffic conditions. Passengers may negotiate within an acceptable range, with the system validating offers and managing the negotiation process until an agreement is reached or the request expires.

Use Case 3: Real-Time Trip Monitoring and Safety Alerts

During a trip, GPS tracking monitors route adherence. If deviations occur, passengers are prompted to confirm safety. Emergency alerts are automatically escalated to administrators when required, enabling timely intervention.

6.2 Data and Process Modelling

The sapU system is designed using a structured analysis approach to support efficient data flow, secure transactions, and real-time monitoring. It functions as a centralized platform

integrating drivers, passengers, administrators, and external services such as payment gateways and GPS providers. Plus, the AI-driven route-matching mechanism is aligned with existing intelligent transportation models that optimize route overlap and minimize detours [3], [4].

Core System Processes

- **User Management:** Handles registration, authentication, profile management, and vehicle details
- **Ride Management:** Manages ride creation, matching, acceptance, updates, cancellations, and completion
- **Navigation & Tracking:** Collects GPS data, validates routes, and detects deviations
- **Payment Processing:** Calculates fares, processes payments, and generates receipts
- **Verification & Rating:** Performs identity checks, driver verification, and rating aggregation
- **Feedback & Reporting:** Manages feedback, disputes, and administrative reporting
- **Emergency & Alert Handling:** Monitors safety events and manages emergency responses

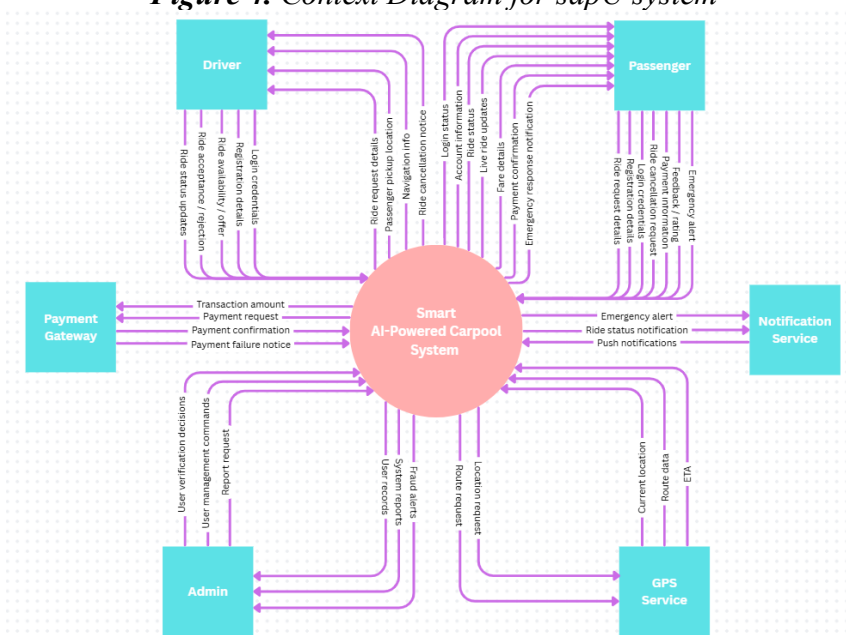
Data Repositories

- **User Database:** Stores user profiles, roles, verification status, and ratings
- **Ride Database:** Maintains ride details, routes, pricing, and status
- **Payment Records:** Tracks all financial transactions and receipts
- **Emergency Records:** Logs safety incidents and administrative responses
- **Feedback Database:** Stores user feedback, ratings, and resolution outcomes

Overall, the analysis demonstrates a clear need for a structured, secure, and data-driven carpooling system. The findings validate sapU's core functionalities and support its role in improving affordability, safety, and efficiency of student transportation.

6.3 DFD Diagrams (Context Diagram, Diagram 0 DFD)

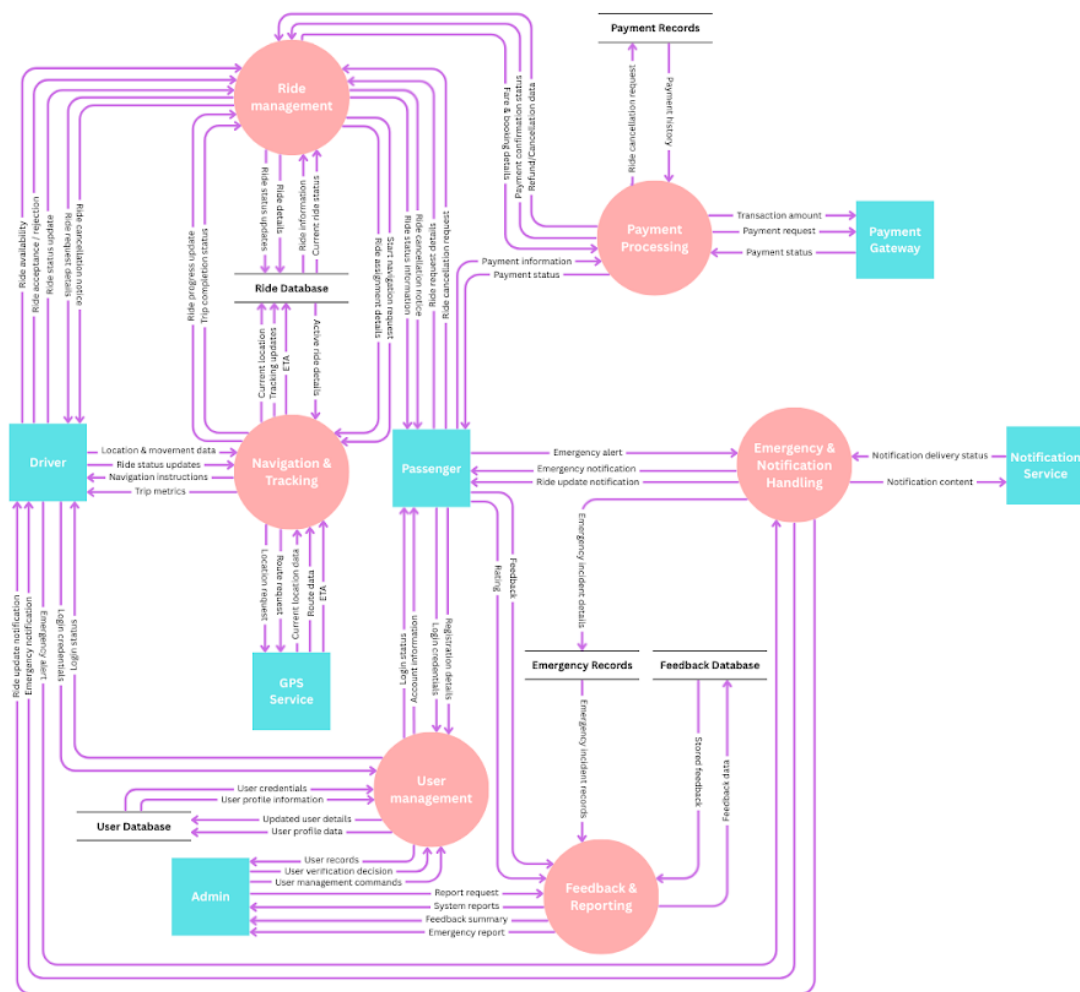
Figure 4. Context Diagram for sapU system



The Context Diagram presents a high-level view of the sapU system and its interactions with external entities. Drivers and passengers exchange ride, location, and payment data with the system, while administrators manage verification, monitoring, and disputes through a dashboard. External services such as the Payment Gateway, Notification Service, and GPS Service support transaction processing, communication, and real-time tracking.

The Context Diagram establishes clear system boundaries and external dependencies, ensuring that sapU functions as a mediator rather than a data owner for payment and location services.

Figure 5. DFD Diagram 0 for sapU system



The Level 0 DFD decomposes the sapU system into seven internal processes. User and ride data flow between User Management, Ride Management, and supporting processes such as Navigation & Tracking and Payment Processing. Safety-related data is handled by the Emergency & Alert Handling process, while Feedback & Reporting supports administrative decision-making.

The decomposition into seven processes improves modularity, scalability, and maintainability, allowing independent enhancement of payment, safety, and navigation components.

6.4 Data Dictionary

The data dictionary defines essential data elements and structures to ensure data consistency across the sapU system.

Core Data Elements

- **user_id, ride_id**: Unique system identifiers
- **email, phone_number, ic_number**: User identity details
- **license_number, vehicle_reg**: Driver and vehicle information
- **pickup_location, dropoff_location**: GPS coordinates
- **fare_amount, platform_fee**: Pricing information
- **ride_status, payment_status, verification_status**: System states
- **rating, user_role**: Trust and access control attributes

Main Data Structures

- **User Profile**: User details, role, verification status, and preferences
- **Driver Profile**: License, vehicle information, and ratings
- **Ride Structure**: Trip details, timing, pricing, and status
- **Payment Transaction**: Financial records and receipts
- **Emergency Incident**: Safety event logs and administrative actions

The data dictionary ensures consistency across processes and supports database normalization by clearly defining data ownership and usage.

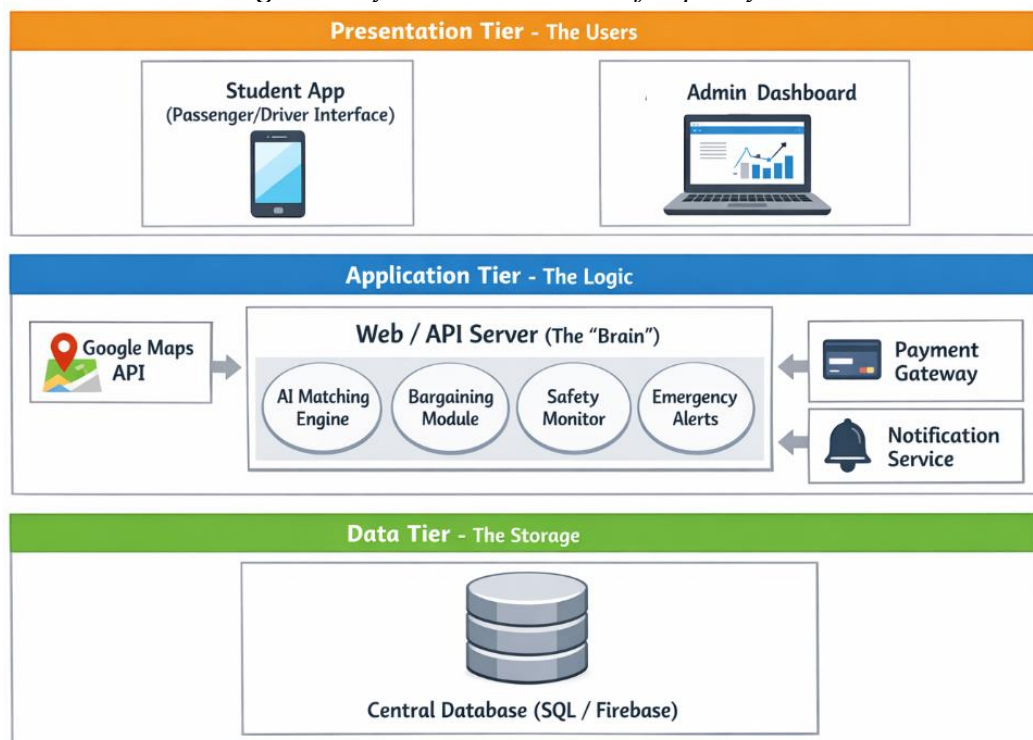
7. DESIGN

7.1 System Architecture

The sapU system follows a Client-Server Architecture designed to handle real-time data processing for GPS tracking and AI route matching. The architecture is divided into three primary layers:

- The Presentation Layer (Mobile/Web App): This is the frontend built for students (Passengers and Drivers) and Admins. It handles user inputs, displays available rides, and provides the interface for the bargaining system.
- The Application Layer (Server/AI Engine): This serves as the "brain" of sapU. It hosts the AI route-matching algorithm that calculates the most efficient paths and coordinates between the GPS API and the notification system.
- The Data Layer (Database): A secure database that stores user profiles, student verification records, ride histories, and payment logs.

Figure 6. System Architecture of sapU System



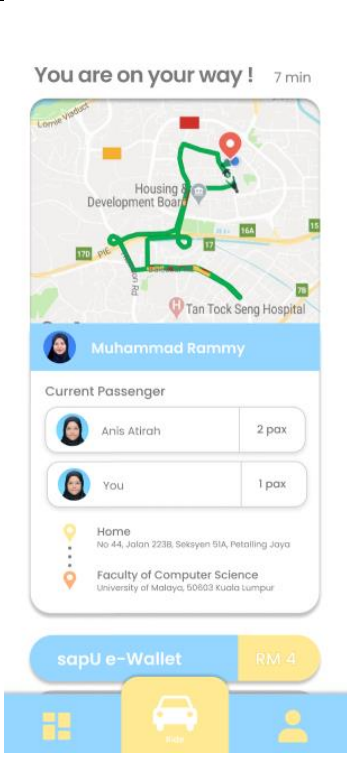
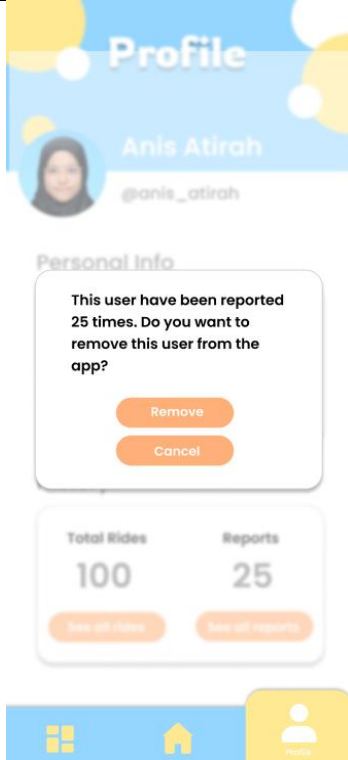
7.2 UI/UX Design Strategy

The design of sapU is centered on **simplicity, speed, and safety**. Because students are often in a rush between classes, the user interface is designed to minimize the number of clicks required to book or offer a ride.

- **User Experience (UX):** We utilize a "Two-Tap Booking" philosophy where frequent routes (e.g., Home to Faculty) are saved for quick access. The navigation flow follows the logic of the Activity Diagram, ensuring a seamless transition from login to ride completion.
- **Visual Design (UI):** The color palette uses trustworthy blues and energetic yellows to reflect both the security and sustainability goals of the carpooling project. High-contrast buttons are used for alert-related actions to ensure they remain highly visible in high-stress situations.

7.3 Wireframes and Key Interfaces

The following wireframes showcase some of the core functionalities of the sapU prototype:

The AI-Matching Dashboard	Real-Time Map View	Admin Verification Portal:
 Customers Nearby Muhammad Batman 4 pax Academy of Malay Studies 50603 Kuala Lumpur, Selangor Kolej Kediaman Ke-9 Jalan 16/8, Seksyen 16, 50603 Petaling Jaya, Wilayah Persekutuan Kuala Lumpur Saya ada bawa dua luggage RM 6 Offer Ride Aqilah Amni Misi 1 pax 24 Hours Study Area, UM Library Lingkungan Budi, 50603 Kuala Lumpur, Wilayah Persekutuan Kuala Lumpur Kolej Kediaman Ke-11 Universiti, 50603 Kuala Lumpur, Federal Territory of Kuala Lumpur No remarks... RM 5 Offer Ride Saharul Ridzwan 3 pax Kolej Kediaman Ke-12 Universiti Malaya, 50603 Kuala Lumpur, Wilayah Persekutuan Kuala Lumpur Ride	 You are on your way! 7 min Muhammad Rammy Current Passenger Anis Atirah 2 pax You 1 pax Home No 44, Jalan 2238, Seksyen 51A, Petaling Jaya Faculty of Computer Science University of Malaya, 50603 Kuala Lumpur sapU e-Wallet RM 4	 Profile Anis Atirah @anis_atirah Personal Info This user have been reported 25 times. Do you want to remove this user from the app? Remove Cancel Total Rides 100 Reports 25 See all rides See all reports

7.4 Database Design

Figure 7. Use Case Diagram of sapU Application

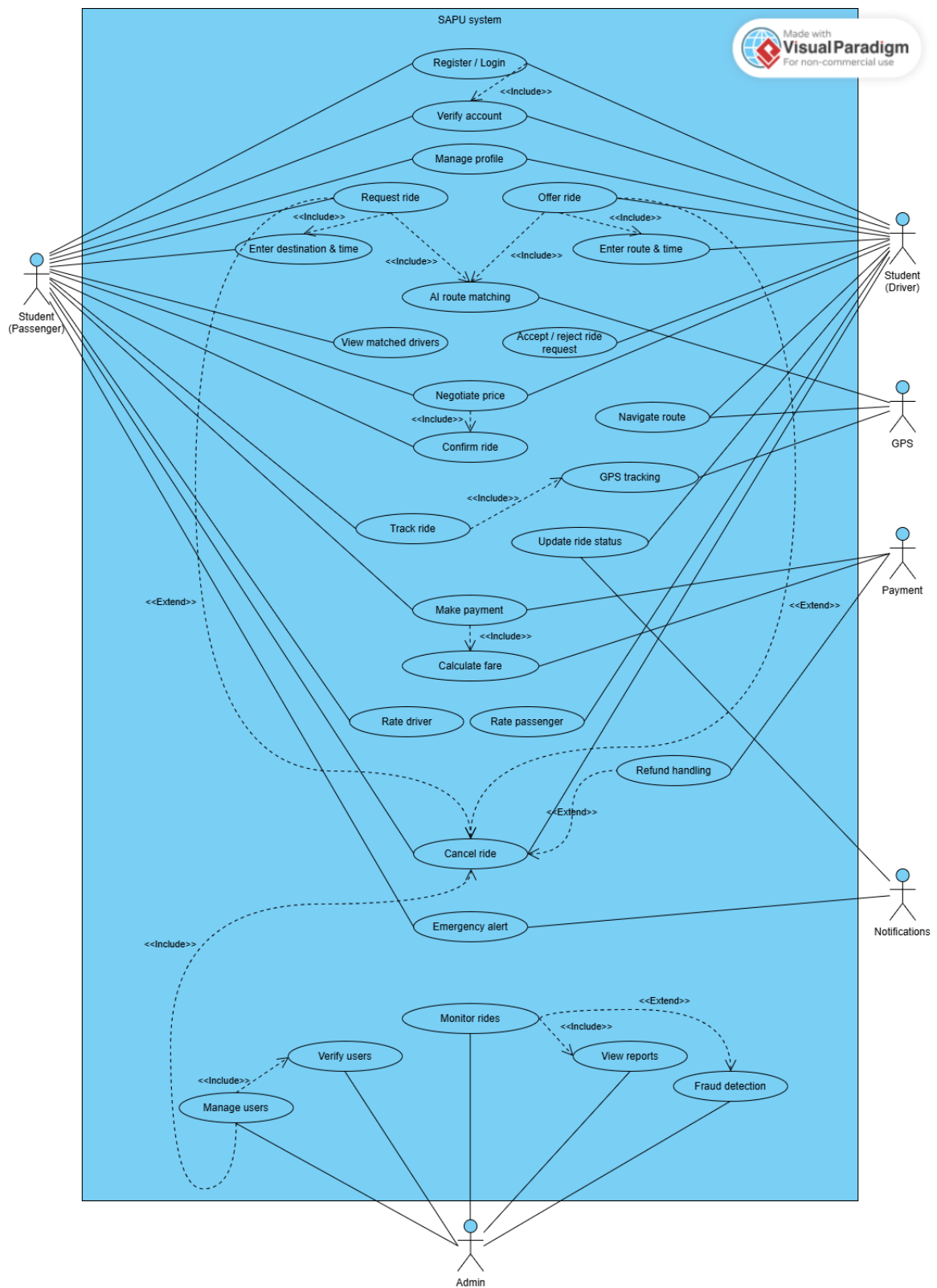
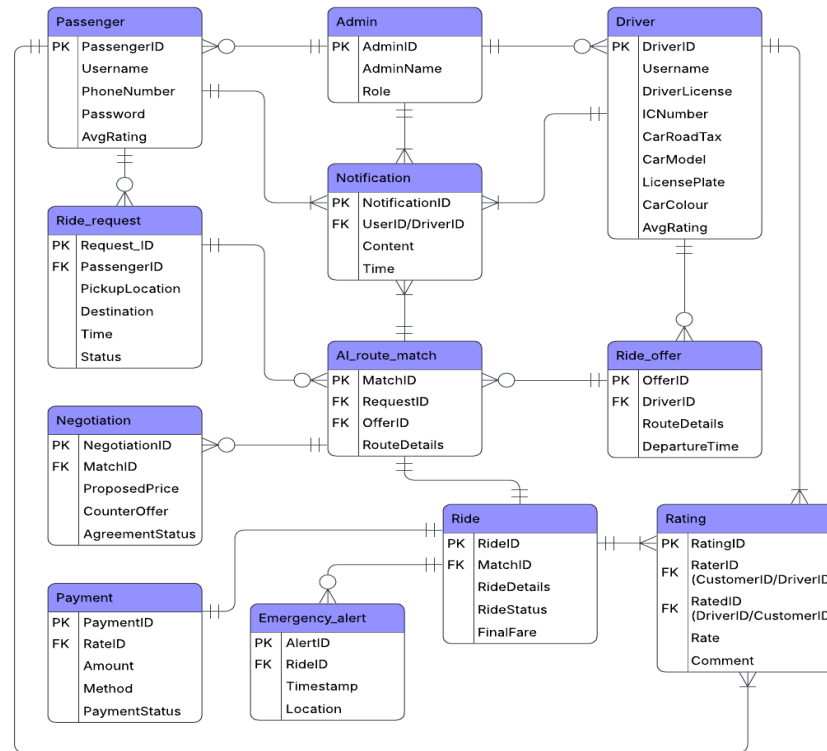


Figure 8. ERD of sapU Application



7.5 Innovative Features

7.5.1 AI-Driven Personalized Matching

The core innovation of sapU is its AI-driven route optimization. Unlike standard apps that simply find the "closest" driver, our algorithm analyzes historical traffic data and overlapping route coordinates to suggest matches that minimize detours for the driver. This ensures that carpooling remains efficient for the driver while staying affordable for the passenger.

7.5.2 Integrated Negotiation Interface

To address the "Financial Stability" problem identified in the planning phase, sapU features a built-in negotiation module. This allows students to counter-offer ride prices within a safe, pre-set range. This feature provides the flexibility of informal WhatsApp groups but with the added security of an official platform.

7.5.3 Real-Time AI Safety Monitoring

Safety is enhanced through an automated monitoring system. If the AI detects a significant deviation from the intended route (via the AI_route_match), it triggers a high-priority Notification to the Admin dashboard and prompts the

student to confirm their safety. An "Emergency Alert" button is also integrated into the UI for immediate assistance.

8. IMPLEMENTATION

8.1 Technical Stack Overview

- a. Frontend (Mobile Application)
 - Platform: Android
 - Technology: Android Studio (Java)
 - Purpose: The purpose of frontend is to provides user interface for drivers and passengers. Other than that, it allows user register, log in, search for rides, offer rides, and manage bookings.
- b. Backend
 - Platform: VSCode
 - Purpose: Backend handles user authentication and authorization for login and sign in pages. Other than that, it manages ride data, user profiles and booking records. Backend also ensures real-time data synchronization between users.
- c. Database
 - Type: Cloud-based Database
 - Purpose: The purpose of database to stores user information, ride details, route data, and transaction history in efficient way. It also supports fast real-time access for mobile users.
- d. AI and Route Matching Module
 - Purpose: AI will match drivers and passengers based on similar routes, locations, and schedules to improve ride efficiency and user satisfaction.
- e. Security and Safety Features
 - Data security: Encrypted cloud storage and access control.
 - Purpose: Security and safety features are important to ensure only authorized users can access the system. It protects user data and enhances trust within the platform.

8.2 Development Methodology

After careful discussion and evaluation, our team has decided to implement the System Development Life Cycle (SDLC) using Agile Methodology for our project. We choose the agile methodology because it provides a flexible and adaptive framework that allows continuous improvement throughout the development process. By using Agile methodology, it allows us to break projects into smaller, iterative, and manageable phases, allowing each stage to be planned and developed effectively. This helps the team stay organized and ensures that every member clearly understands their individual tasks and responsibilities.

Moreover, the Agile methodology fosters consistent communication and collaboration among team members, facilitating the early identification and resolution of issues and challenges. This helps the team to continuously enhance the system based on feedback from users or

stakeholders. This will enhance the overall product quality, ensuring the outcome aligns with the user's needs and expectations.

In addition, the Agile approach has the potential to lower project costs and risks. Each iteration involves testing and evaluation, allowing for the early detection and correction of potential problems before they escalate into major issues. This strategy prevents unnecessary work and delays and thus can save time and resources. In conclusion, the Agile methodology within the SDLC framework provides an efficient, flexible, and risk-reducing approach.

8.3 Implementation - Screen shots of User Interface Design based on diagrams.

- Login and Sign-Up Feature



Figure 9.1.1 Welcome Page

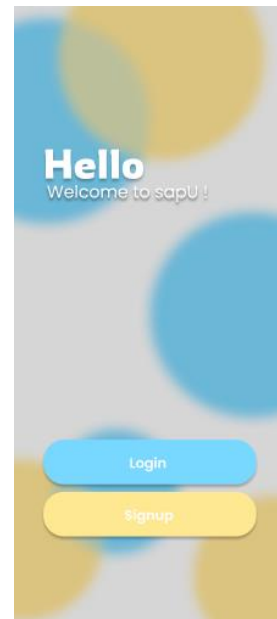


Figure 9.1.2 Welcome Page 2

Login

Customer Driver Admin

Username

Password

Login

Figure 9.1.2 Login Page

Signup

Customer Driver

Name

Birth Date

Phone Number

Email

Set username

Set password

By signing up, you agree on our Terms and Privacy.

Signup

Figure 9.1.3 Sign-Up Page

- Customer Feature

Where to go?

Choose Destination

Faculty of Computer Science
University of Malaysia, 50603 Kuala Lumpur

Bangunan Peperiksaan UM
University of Malaysia, 50603 Kuala Lumpur

Choose Pickup

Current Location

Home
No 44, Jalan 2238, Seksyen 51A, Petaling Jaya

UM Central Point
University of Malaysia, 50603 Kuala Lumpur

Number of pax: 1

Search Ride

Ride

Figure 9.2.1 Ride Page

Available Rides

Muhamad Rammy 5 min

Current Passenger

Anis Atrah 2 pax

Faculty of Computer... Select Ride

Fatin Nurliyana 7 min

Current Passenger

No passenger

Faculty of Computer... Select Ride

Lee Minho 10 min

Current Passenger

Fatin Nurliyana 2 pax

Ride

Figure 9.2.2 Available Rider Page

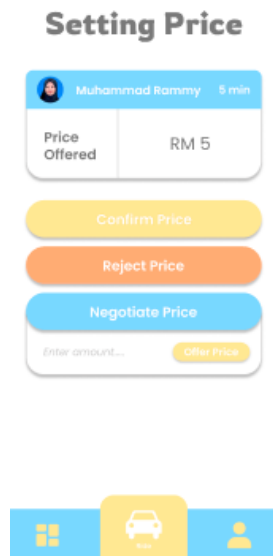


Figure 9.2.3 Setting Price Page

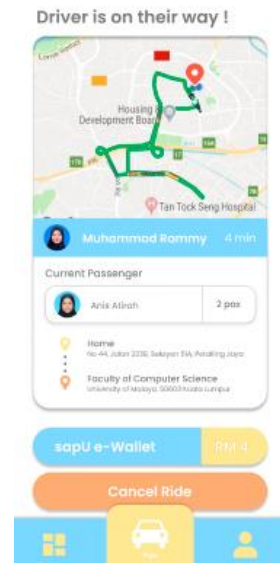


Figure 9.2.4 Ongoing Ride Page

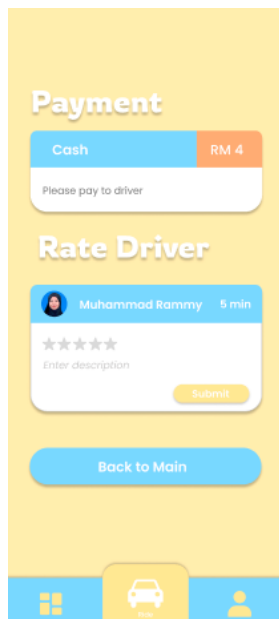


Figure 9.2.5 Payment Option (Cash)

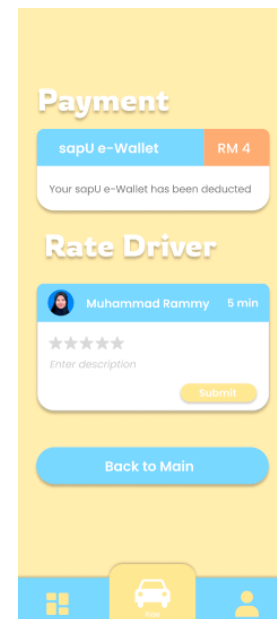


Figure 9.2.6 Payment Option (e-Wallet)

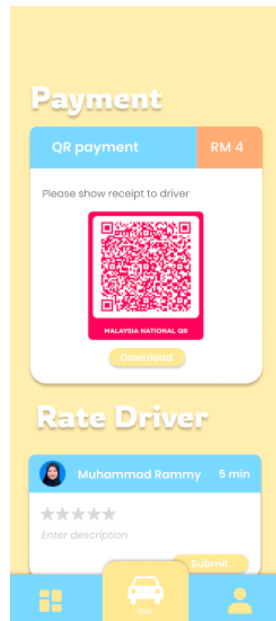


Figure 9.2.7 Payment Option (Payment)



Figure 9.2.8 Profile Page



Figure 9.2.9 Dashboard Page

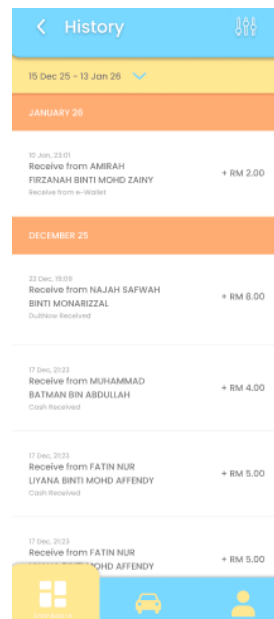


Figure 9.2.10 Transaction History Page

<

Emergency Helpline

What is your emergency?

Accident

Medical Emergency

Harrasment or Assault

Vehicle Breakdown

Physical violence or threats

Environment Emergency

Others...

Send Alert

Emergency Helpline

Figure 9.2.11 Emergency Helpline Page

<

Fraud Report

Please describe your situation.

State here

Send Report

Fraud Report

Figure 9.2.12 Fraud Report Page

- Driver Feature



Figure 9.3.1 Customer Nearby Page

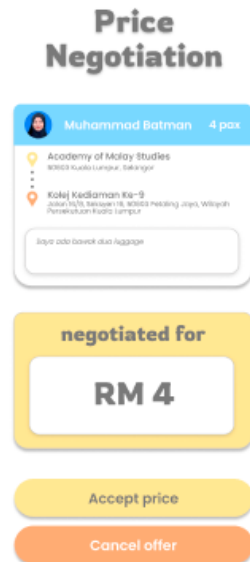


Figure 9.3.2 Negotiation Page

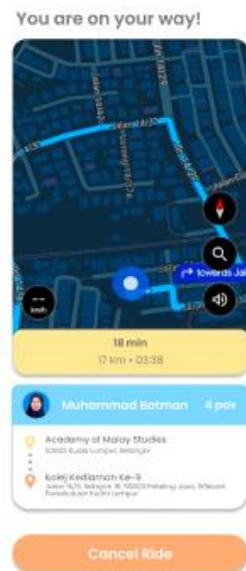


Figure 9.3.3 Navigation Page

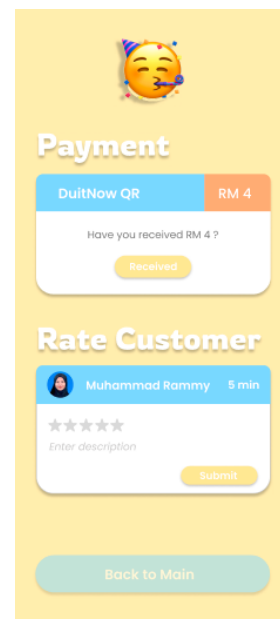


Figure 9.3.4 Payment Received via DuitNow

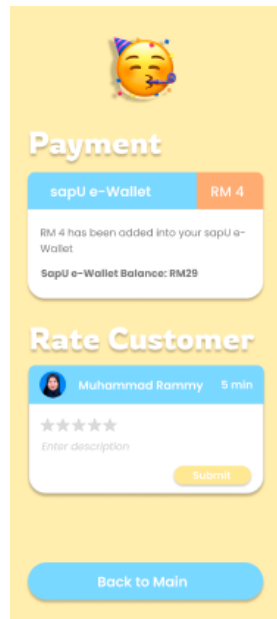


Figure 9.3.5 Payment Received via e-Wallet

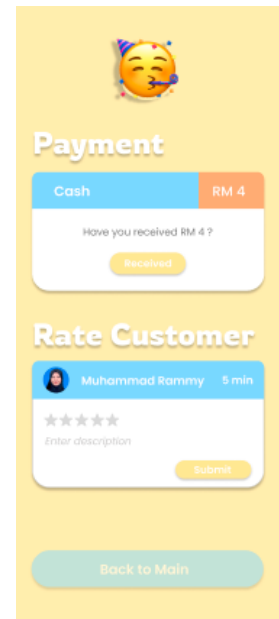


Figure 9.3.6 Payment Received Via Cash

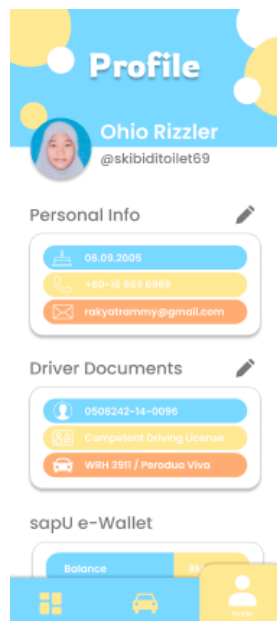


Figure 9.3.7 Profile Page

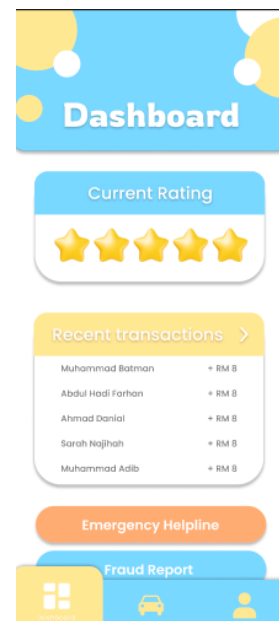


Figure 9.3.8 Dashboard Page

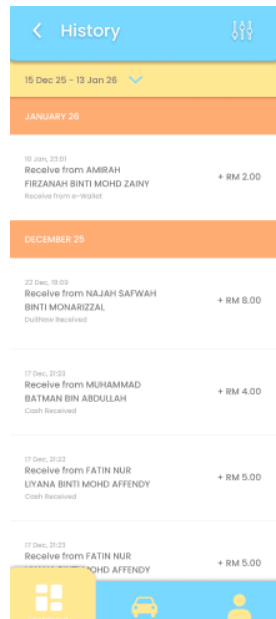


Figure 9.3.9 Transaction History

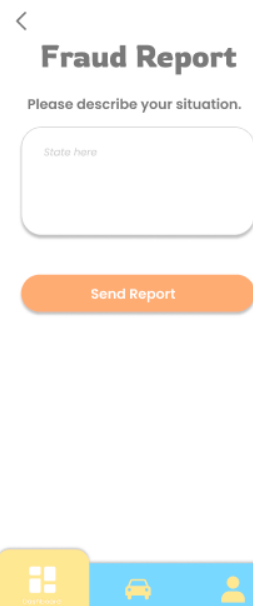


Figure 9.3.10 Fraud Report Page

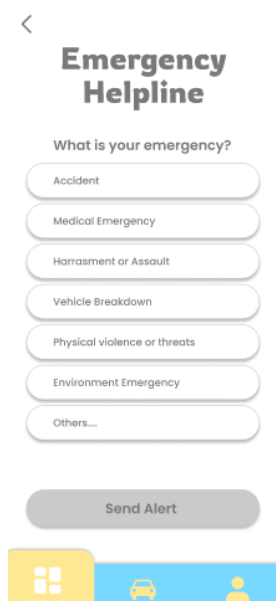


Figure 9.3.11 Emergency Helpline Page

- Admin Feature

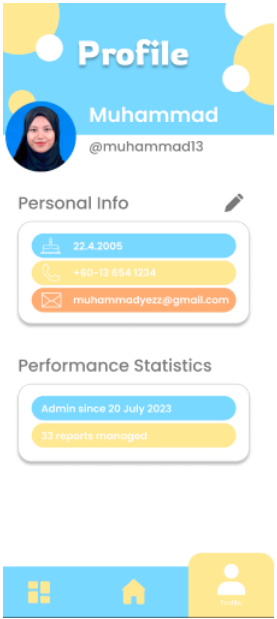


Figure 9.4.1 Admin Profile Page

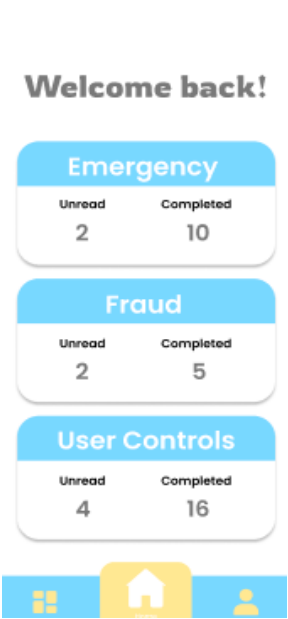


Figure 9.4.2 Admin Home Page

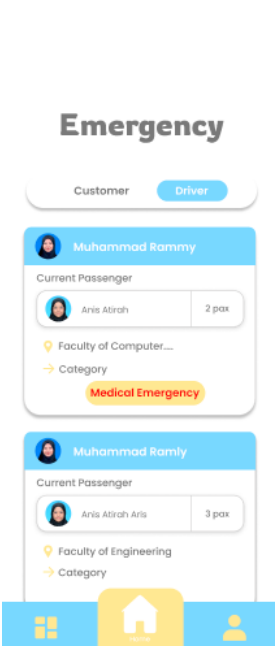


Figure 9.4.3 Emergency Page

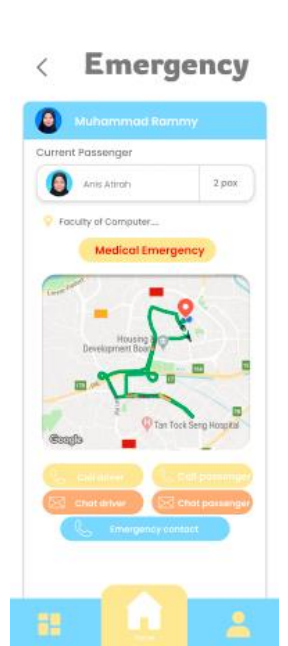


Figure 9.4.4 Emergency Details Page

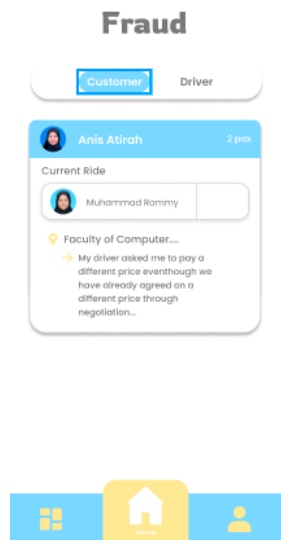


Figure 9.4.5 Fraud Page

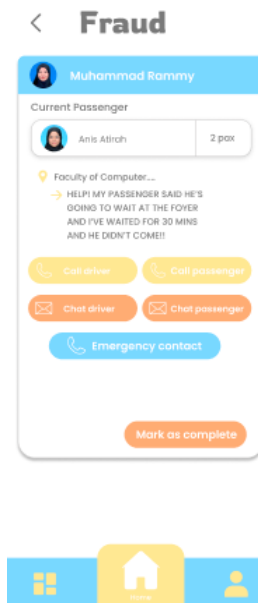


Figure 9.4.6 Fraud Details Page



Figure 9.4.7 User Controls Page

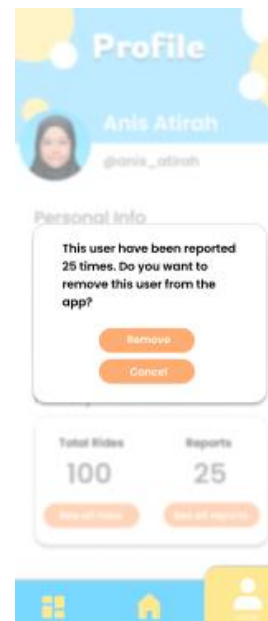


Figure 9.4.8 Reported User Profile

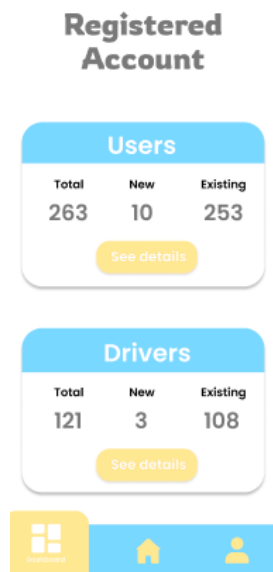


Figure 9.4.9 Admin Dashboard Page

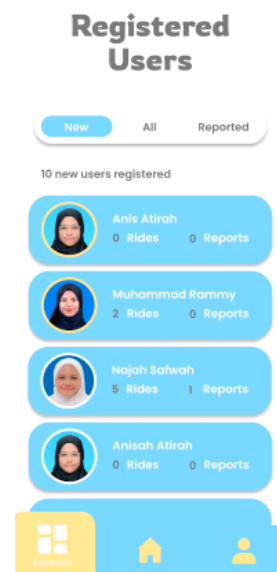


Figure 9.4.10 Registered User Page

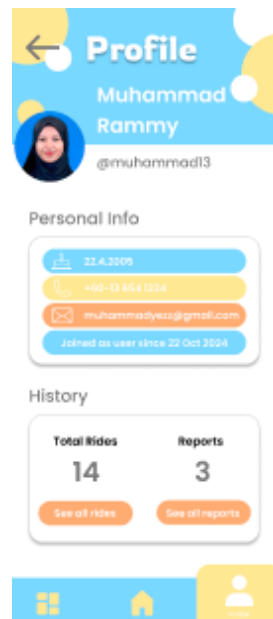


Figure 9.4.11 User Profile Info Page

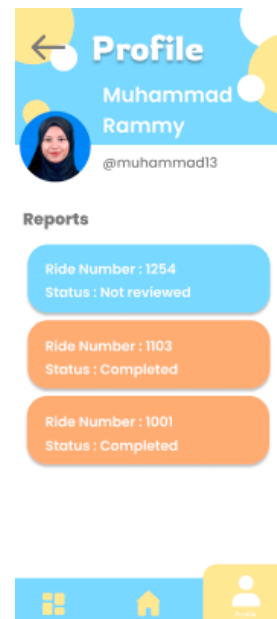
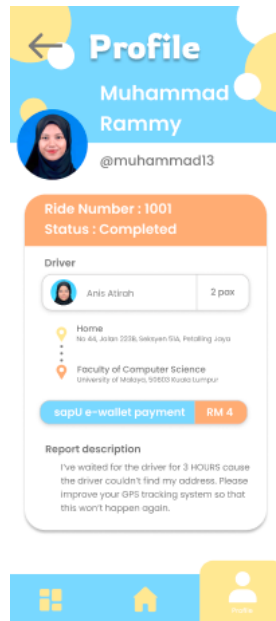


Figure 9.4.12 User Report List Page



*Figure 8.12 User Report Details
(Complete Ride)*



*Figure 8.13 User Report Details
(Ongoing Ride)*

9. EVALUATION AND TESTING

The section outlines the evaluation and testing procedures carried out to ensure the sapU (AI-driven) operates effectively and aligns with its requirements, standards, and users' expectations. The evaluation and testing process focuses on verifying system reliability, usability, and overall performance. To achieve this, multiple testing techniques will be done, including seven types of testing, usability testing, feedback incorporation, and user acceptance testing.

9.1 Overview of System Evaluation and Testing

System evaluation and testing will be carried out throughout the development of the project to identify the error early and to improve system quality. Both static and dynamic testing methods were applied to ensure this application functions effectively and is ready for real-world scenarios. These activities were aligned with the Agile development methodology that we adopted that allows continuous evaluation and improvement.

9.2 Types of Testing

Seven types of testing will be conducted to evaluate different aspects of the sapU application.

- **Inspection**
Inspection will be conducted by reviewing system documentation, design, and source code without executing the system. This testing helps identify the consistency of the system, requirements, and logical errors at an early stage.
- **Walk-through**
Walk-through sessions are where developers explained system logic and workflows with team members. At this session, feedback and suggestions will be collected to improve system design and functionality.
- **Desk Checking**
Developers will manually trace program logic before code execution. This method will help ensure the system's correctness by detecting logical errors early in the development process.
- **Unit Testing**
This focuses on testing individual system components independently. Functions such as user registration, login validation, ride creation and booking confirmation will be tested to make them function well.
- **Integration Testing**
This testing will verify that different system modules worked correctly when combined to ensure smooth communication between the frontend, backend, and database.
- **System Testing**
System testing evaluates the complete system for real-world usage scenarios. This is to make sure the system meets all the requirements and is ready to use.

- **Stub Testing**

The goal of stub testing is to ensure that the component being tested is isolated from external factors, making sure that the test results are reliable, predictable, and faster.

9.3 Plan for Usability Testing

The usability testing objective is to evaluate how users can interact with the system and complete common tasks such as signing in as a user, booking a ride and matching with a rider. This is to ensure that the system interface is user-friendly and supports efficient task completion.

Selected students or participants will be chosen to represent drivers and passengers. They will perform tasks such as registration, login, booking or accepting a ride. Then, user interactions will be observed, and feedback will be collected through questionnaires and interviews.

9.4 Feedback Incorporation

User feedback will be collected during the usability testing stage and system testing. Then, the feedback will be analyzed and categorized based on usability, interface design, and performance to help with further improvements.

9.5 User Acceptance Testing

User acceptance testing will be conducted using beta testing approaches. Selected users from the target user group, which is students, will test the sapU system in real time. The purpose of beta testing is to evaluate whether the system meets user expectations, supports real-world usage and is ready to deploy. Feedback will be collected to make final improvements before system release.

9.6 Summary

In conclusion, the evaluation and testing phase ensures that the sapU system is reliable, functional, and user-friendly. By applying seven types of testing, conducting usability testing, incorporating user feedback, and performing user acceptance testing, the system was refined to meet both technical requirements. These processes confirm that the sapU system is suitable for deployment as a smart carpooling platform.

10. CONCLUSION

Throughout the development of the sapU (Smart AI-Powered Carpool System), the project team gained a comprehensive understanding of the transportation challenges faced by university students and the ways technology can address these issues. The study highlighted critical insights into student mobility, including the financial strain of commercial e-hailing services, the operational inefficiencies of informal ridesharing, and the safety concerns associated with unregulated platforms. The project demonstrated that AI-driven route-matching, integrated bargaining, and real-time safety monitoring can significantly enhance the commuting experience while promoting sustainability by reducing campus traffic and carbon emissions.

Despite these achievements, further research is necessary to optimize the AI matching algorithm for increasingly complex traffic patterns and to explore expansion opportunities beyond campus boundaries. Additionally, integrating predictive analytics could improve ride availability forecasting and enhance user experience. The project also faced limitations, including a constrained budget, reliance on third-party GPS data, and the exclusion of long-distance rides, which may restrict scalability and real-world applicability in broader contexts.

The sapU system offers clear benefits and practical applications. Students can enjoy safer, more cost-effective, and environmentally friendly commuting options, while student-drivers gain supplementary income opportunities. The structured platform reduces redundancies, increases reliability, and builds trust through verified user interactions. Based on the findings, it is recommended to continue refining the AI matching and safety features, explore multi-campus deployment, and periodically incorporate user feedback to maintain alignment with student needs. Overall, the sapU project successfully demonstrates the potential of an AI-powered, student-centered carpooling system to transform campus transportation.

REFERENCES

Sources: Journal, Technical report, Magazine, Book, Bulletin, Thesis, internet source (please do not use Wikipedia)

Style: IEEE referencing and citation format (Choose “Chicago” when you click “cite”)

Databases: Google Scholar <https://scholar.google.com/> (Custom range 2020 – 2025) and others

[1] A. Shaheen and N. Chan, “Mobility and the Sharing Economy: Potential to Facilitate the First- and Last-Mile Public Transit Connections,” *Built Environment*, vol. 42, no. 4, pp. 573–588, 2020.

[2] S. Acheampong et al., “Students’ travel behaviour and sustainable transport choices in university campuses,” *Transportation Research Part D*, vol. 86, 2020.

[3] Y. Wang et al., “AI-based route matching for ride-sharing services using real-time traffic data,” *IEEE Access*, vol. 9, pp. 102345–102356, 2021.

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APPENDIX

Appendix A : Survey Questionnaires

Note: This questionnaire was distributed to students of the University of Malaya to understand their transportation patterns and challenges within the campus.

SECTION A: Demographic Information

1. What is your age?
 - ☐ 20
 - ☐ 20-24
 - ☐ Above 24
2. Which year are you in?
 - ☐ Year 1
 - ☐ Year 2
 - ☐ Year 3
 - ☐ Year 4
3. Do you live on campus grounds or outside?
 - ☐ Yes, I live on campus grounds
 - ☐ No, I live outside of UM

SECTION B: Transportation Usage

4. Do you have your own transport?
 - ☐ Yes
 - ☐ No
5. If you do, please share with us your mode of transportation.
 - ☐ Car
 - ☐ Motorcycle
 - ☐ Bicycle
 - ☐ Others
6. If you don't, please share with us your mode of transportation.
 - ☐ Grab/E-hailing
 - ☐ Public transport
 - ☐ Carpool
 - ☐ Walking
 - ☐ Others

Appendix B : Interview Questionnaires and Responses

Note : Three interview responses were collected for this survey.

[Respondent A, B and C]

1. On a typical week, how much do you spend on e-hailing services like Grab or AirAsia, and how does this expenditure affect your overall financial stability?

A - On a typical week, I spend around RM40–RM70 on Grab. It affects my finances because I need to save money as a student.

B - Rm 5, I only use e-hailing service in case of emergency, other than that I use free bus ride.

C - Zero. I own a car. I do spend roughly RM100 per week on tolls & gas.

2. When using informal ride-sharing methods like WhatsApp or Telegram groups, what are your primary safety concerns regarding the lack of GPS tracking or user verification?

A - My main concerns are safety, because there is no GPS tracking, no proper driver verification, and it is hard to trace issues if something happens.

B - Some people can be quite suspicious especially for women passengers.

C - Not knowing the driver's ETA to properly plan my journey. I often arrive late to my destination because of this. I'm also concerned about their identity.

3. How much time do you typically waste waiting for a match?

A - I usually waste around 10–20 minutes waiting for a match, especially during peak hours.

B - Less than 3 min

C - 10 minutes~

4. If a system allowed you to negotiate a ride price directly with a driver who is already heading in your direction, how likely would you be to switch from your current transport method?

A - I am very likely to switch because it would be cheaper and more flexible.

B - Yes, sure.

C - No, I would stick to my current transport method.

5. If you own a vehicle, would a structured AI system that matches your exact route encourage you to earn side income by offering rides to fellow students?"

A - Yes, an AI system like that would encourage me to earn side income because it feels more organized and safer.

B - Yes, I would.

C - Yes, I would, as it would reduce my financial burden on gas & tolls.

Appendix C : Interview Session



Interview with Respondent A



Interview with Respondent B